University of Ottawa

Department of Mathematics and Statistics

MAT 1302C : Mathematical Methods II

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Second Midterm Exam – Version A

November 22, 2012

Surname _____ First Name _____

Student # _____ DGD (Tuesday/Thursday) _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(d) Write your student number at the top of each page in the space provided.
(e) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(f) You are strongly recommended to write in **pen**, not pencil.
(g) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	5	3	4	3	4	5	24
Grade							

1. Let

$$A = \begin{bmatrix} 1 & -3 & 2 & -4 \\ -3 & 10 & -1 & 5 \\ 2 & -5 & 4 & -3 \\ -4 & 12 & 2 & -3 \end{bmatrix}.$$

(a) **[2.5 points]** Calculate $\det A$ by using the row reduction method.

Solution:

$$\det A = \begin{vmatrix} 1 & -3 & 2 & -4 \\ -3 & 10 & -1 & 5 \\ 2 & -5 & 4 & -3 \\ -4 & 12 & 2 & -3 \end{vmatrix} \overset{\substack{R_2\to R_2+3R_1 \\ R_3\to R_3-2R_1 \\ R_4\to R_4+4R_1}}{=} \begin{vmatrix} 1 & -3 & 2 & -4 \\ 0 & 1 & 5 & -7 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 10 & -19 \end{vmatrix}$$

$$\overset{R_3\to R_3-R_2}{=} \begin{vmatrix} 1 & -3 & 2 & -4 \\ 0 & 1 & 5 & -7 \\ 0 & 0 & -5 & 12 \\ 0 & 0 & 10 & -19 \end{vmatrix} \overset{R_4\to R_4+2R_3}{=} \begin{vmatrix} 1 & -3 & 2 & -4 \\ 0 & 1 & 5 & -7 \\ 0 & 0 & -5 & 12 \\ 0 & 0 & 0 & 5 \end{vmatrix}$$

$$= 1(1)(-5)5 = -25.$$

(b) **[1.5 points]** Find C_{23}, the (2,3)-cofactor of A.

Solution:

$$C_{23} = (-1)^{2+3} \det A_{23} = -\det \begin{bmatrix} 1 & -3 & -4 \\ 2 & -5 & -3 \\ -4 & 12 & -3 \end{bmatrix}$$

$$\overset{\substack{R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 + 4R_1}}{=} - \begin{vmatrix} 1 & -3 & -4 \\ 0 & 1 & 5 \\ 0 & 0 & -19 \end{vmatrix} = -1(1)(-19) = 19.$$

(c) **[1 point]** (This part is independent of (a) and (b) above.)

Suppose A is an $n \times n$ invertible matrix. Which of the following statements is **false**? Circle the correct answer.

(a) A^{-1} is invertible.
(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
(c) If B is invertible then AB is invertible.
(d) A^T is invertible.
(e) The determinant of A is 0.

Solution: (e).

2. Determine if the following sets are subspaces of $\mathbb{R}^3$ or not. Justify your answer.
(a) **[1.5 points]**

$$U = \left\{ \begin{bmatrix} a-b \\ a+b \\ 3a \end{bmatrix} \middle| a,\, b \in \mathbb{R} \right\}$$

Solution: We show that U is a span of a given set of vectors:
For $\begin{bmatrix} a-b \\ a+b \\ 3a \end{bmatrix} \in U$,

$$\begin{bmatrix} a-b \\ a+b \\ 3a \end{bmatrix} = \begin{bmatrix} a \\ a \\ 3a \end{bmatrix} + \begin{bmatrix} -b \\ b \\ 0 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

with $a,\, b \in \mathbb{R}$. Therefore $U = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$ and hence is a subspace of $\mathbb{R}^3$.

(b) **[1.5 points]**

$$V = \left\{ \begin{bmatrix} a \\ b \\ ab \end{bmatrix} \middle| a,\, b \in \mathbb{R} \right\}$$

Solution: V is not a subspace of $\mathbb{R}^3$ for the following reasons:

(a) V is not closed under scalar multiplication. For example, $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \in V$, but

$$2 \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} \notin V,$$

since the third entry is not the multiple of the first and second entries.

(b) V is not closed under addition. For example, $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \in V$, but

$$\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix} \notin V,$$

since the third entry is not the multiple of the first and second entries.

3. Consider the matrix

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 3 & 11 & 2 \end{bmatrix}.$$

(a) **[3 points]** Find the inverse of the matrix A.

Solution:

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 2 & 5 & 2 & 0 & 1 & 0 \\ 3 & 11 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & 2 & -1 & -3 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & 0 & -1 & -7 & 2 & 1 \end{array}\right] \xrightarrow{R_1 \to R_1 + R_3} \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & -6 & 2 & 1 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & 0 & -1 & -7 & 2 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 + 3R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -12 & 5 & 1 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & 0 & -1 & -7 & 2 & 1 \end{array}\right] \xrightarrow[R_3 \to -R_3]{R_2 \to -R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -12 & 5 & 1 \\ 0 & 1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & 7 & -2 & -1 \end{array}\right]$$

Hence, A is invertible and $A^{-1} = \begin{bmatrix} -12 & 5 & 1 \\ 2 & -1 & 0 \\ 7 & -2 & -1 \end{bmatrix}$

(b) **[1 point]** Use A^{-1} to solve $A\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

Solution: $\mathbf{x} = A^{-1} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -12 & 5 & 1 \\ 2 & -1 & 0 \\ 7 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -11 \\ 2 \\ 6 \end{bmatrix}$.

4. (a) **[1.5 points]** Solve the matrix equation

$$BX^TA^{-1} + CB = 0,$$

for X in terms of A, B, C. Here A, B, C and X are 2×2 invertible matrices.

Solution:

- Move CB to the right hand side::
$$BX^TA^{-1} = -CB,$$
- Multiply from the left by B^{-1} and from the right by A:
$$X^T = -B^{-1}CBA,$$
- Take transpose on two sides:
$$X = (-B^{-1}CBA)^T = -A^TB^TC^T(B^{-1})^T.$$

(b) **[1.5 points]** Let A, B and C be 2×2 matrices with

$$\det A = 6, \quad \det B = -3, \quad \det\left(2A^TB^{-1}C\right) = 40.$$

Compute $\det C$.

Solution:

$$40 = \det\left(2A^TB^{-1}C\right) = 2^2(\det A^T)(\det B^{-1})(\det C)$$

$$= 4(\det A)(\frac{1}{\det B})(\det C) = 4(6)(\frac{1}{-3})(\det C) = -8(\det C)$$

Thus $\det C = \frac{40}{-8} = -5$.

5. Let $A = \begin{bmatrix} 1 & 0 & -2 & -4 \\ -1 & 1 & -1 & 4 \\ -2 & -2 & 10 & 8 \end{bmatrix}$. The matrix $\begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is the row reduced echelon form of A.

(a) [**2 points**] Find a basis for Col A. What is rank A? **Solution:** The pivot columns of A are the first, and the second. Therefore a basis for Col A is given by

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right\}.$$

rank A = number of pivot columns of $A = 2$.

(b) [**2 points**] Find a basis for NulA.

Solution: To find a basis for Nul A, we should write the general solution of $A\mathbf{x} = \mathbf{0}$ in parametric form. From the given reduced form of A we have $x_1 = 2x_3 + 4x_4$, $x_2 = 3x_3$, x_3 and x_4 are free. The solution of $A\vec{x} = \vec{0}$ is

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2x_3 + 4x_4 \\ 3x_3 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2x_3 \\ 3x_3 \\ x_3 \\ 0 \end{bmatrix} + \begin{bmatrix} 4x_4 \\ 0 \\ 0 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 4 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

A basis for NulA is $\left\{ \begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$.

6. An economy consists of two sectors: Agriculture and Construction. For each unit of output, Agriculture requires 0.50 units from Agriculture and 0.70 units from Construction. For each unit of output, Construction uses 0.40 units from Agriculture and 0.40 units from Construction.

(a) **[1 point]** What is the consumption matrix C for this economy?

(b) **[2 points]** Find $I - C$ and $(I - C)^{-1}$.

(c) **[1 point]** Determine what intermediate demands are created if Agriculture plans to produce 100 units.

(d) **[1 point]** Find the production levels that will satisfy the final demand of 100 units from Agriculture and 200 units from Construction.

Solution: (a) The consumption matrix of this economy is:

$$C = \begin{bmatrix} 0.5 & 0.4 \\ 0.7 & 0.4 \end{bmatrix}.$$

(b)

$$I - C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.5 & 0.4 \\ 0.7 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.4 \\ -0.7 & 0.6 \end{bmatrix}.$$

$$(I - C)^{-1} = \frac{1}{0.5(0.6) - (-0.4)(-0.7)} \begin{bmatrix} 0.6 & 0.4 \\ 0.7 & 0.5 \end{bmatrix} = \frac{1}{0.02} \begin{bmatrix} 0.6 & 0.4 \\ 0.7 & 0.5 \end{bmatrix} = \begin{bmatrix} 30 & 20 \\ 35 & 25 \end{bmatrix}.$$

(c)

$$100 \begin{bmatrix} 0.5 \\ 0.7 \end{bmatrix} = \begin{bmatrix} 50 \\ 70 \end{bmatrix}.$$

(d) Let x_1 and x_2 be the outputs of Agriculture and Construction respectively. Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be the production vector. Note that $\mathbf{d} = \begin{bmatrix} 100 \\ 200 \end{bmatrix}$.

$$\mathbf{x} = (\mathbf{I} - \mathbf{C})^{-1}\mathbf{d} = \begin{bmatrix} 30 & 20 \\ 35 & 25 \end{bmatrix} \begin{bmatrix} 100 \\ 200 \end{bmatrix} = \begin{bmatrix} 7000 \\ 8500 \end{bmatrix}.$$